

Neoplants Market Report: Bioengineered Air Purification

Executive Summary

Neoplants operates in the emerging "nature tech" sector, creating bioengineered plants for indoor air purification with their flagship Neo Px product priced at \$179. The company claims their system is 30x more effective than regular houseplants at removing harmful VOCs (Benzene, Toluene, Xylene) and covers 160 square feet. Neoplants has raised \$20M in funding and competes with companies like Ubreathе in the plant-based air purification market. The company reportedly generates \$7.4M in annual revenue with 48 employees. Their approach combines biotechnology with traditional houseplants, targeting health-conscious consumers concerned about indoor air quality through direct-to-consumer sales.

Quick Answers

What can be found on it? Neo Px system includes: bioengineered Power Drops (proprietary microorganisms), custom PLA planter with water reservoir, and Marble Queen Pothos plant. System covers 160 sqft and operates 24/7 at 0dB.

What is it good for? Targets the three most harmful indoor VOCs (Benzene, Toluene, Xylene) with 30x efficiency vs regular plants. Reduces toxic chemicals by 30% within 7 days. 86% of users report wellbeing improvements.

Why do we need it? Traditional air purifiers are inefficient against dangerous VOCs. Neo Px captures and recycles these pollutants into organic matter, providing continuous air purification in homes where people spend 90% of their time.

Deep Dive

Feature	Details	Source
Product Type	Bioengineered plant air purification system	Neoplants
Core Technology	Power Drops - enhanced Pseudomonas putida bacteria via directed evolution	Neoplants
Coverage Area	160 square feet per unit	Neoplants
Target Pollutants	Benzene, Toluene, Xylene (BTX compounds)	Neoplants
Effectiveness Claim	30x more effective than regular houseplants	Neoplants
Operating Specs	0 dB, 0 Watts, 24/7 operation	Neoplants

Business Metrics	Details	Source
Funding Raised	\$20M across 3 rounds (latest: Seed VC Oct 2022)	CB Insights
Annual Revenue	\$7.4M (estimated)	Growjo
Employee Count	48 employees	Growjo
Key Investors	True Ventures, Collaborative Fund, Partech, Heartcore Capital	CB Insights

Pros & Cons

Pros	Cons	Source
30x more effective than regular plants	Requires monthly Power Drops refills (\$9/month ongoing cost)	Neoplants
Zero energy consumption (0 Watts)	Limited to 160 sqft coverage area	Neoplants
Silent operation (0 dB)	High initial cost (\$179) vs traditional plants	AgFunder
86% user satisfaction for wellbeing	Requires specific care conditions (humidity 50-70%)	Neoplants
Sustainable materials (biodegradable PLA shell)	Limited pollutant scope (only BTX compounds)	Neoplants

Competitors

Product	Company	Description	Standout Features	Pricing	Link/Contact
Breathing Roots Technology	Ubreathe	Nature-based air purification systems	Patented breathing roots technology	Not disclosed	CB Insights
AlgenAir	AlgenAir	Algae-based air purification	Uses algae instead of plants	Not disclosed	CB Insights
Bloomscape	Bloomscape	Traditional houseplant delivery service	Wide plant variety, care guidance	\$35-200+	CB Insights

Comparison Matrix

Table 1 – Basics

Product	Company	Form Factor	Best For	Links
Neo Px	Neoplants	Bioengineered plant + planter	VOC removal, small spaces (160 sqft)	neoplants.com
Traditional Air Purifiers	Various	Electric HEPA filters	Particle filtration, larger spaces	Retail stores
Regular Houseplants	Various	Natural plants	Basic air improvement, aesthetics	Garden centers

Table 2 – Capabilities

Product	Pollutant Targets	Effectiveness Claim	Processing Method
Neo Px	Benzene, Toluene, Xylene	30x vs regular plants	Biological conversion to organic matter
HEPA Purifiers	Particles, some VOCs	CADR ratings vary	Physical filtration
Regular Plants	General air quality	NASA studies (limited VOC removal)	Natural plant processes

Table 3 – Ecosystem

Product	Maintenance	Data Controls
Neo Px	Monthly Power Drops refills, watering every 2-3 weeks	No data collection mentioned
Electric Purifiers	Filter replacements, cleaning	App-connected models may collect data
Regular Plants	Watering, fertilizing, pruning	No data involved

Table 4 – Commercial

Product	Pricing	Ongoing Costs	Limitations
Neo Px	\$179 initial	\$9/month Power Drops	160 sqft coverage, monthly maintenance
HEPA Purifiers	\$100-800+	\$20-100/year filters	Energy consumption, noise
Regular Plants	\$10-100+	Minimal (soil, fertilizer)	Limited air purification effectiveness

Decision Guide

By Persona:

- **Health-conscious homeowners:** Neo Px ideal for targeted VOC removal in bedrooms/offices
- **Apartment dwellers:** Good fit for small spaces (160 sqft coverage), silent operation
- **Tech enthusiasts:** Appeals to those wanting cutting-edge biotechnology solutions
- **Plant lovers:** Combines air purification with traditional houseplant aesthetics
- **Budget-conscious consumers:** Consider ongoing \$108/year Power Drops cost vs alternatives

Red Flags:

- **Regulatory concerns:** Bioengineered organisms may face future regulations
- **Scalability:** Limited coverage area requires multiple units for larger homes
- **Dependency:** System effectiveness relies on continued Power Drops purchases
- **Scientific validation:** Claims based on lab studies, limited independent verification

Implementation Tips

Scenario	Tip	Why It Matters
Small apartment	Place in bedroom or home office	Maximizes health benefits where you spend most time
Multiple rooms	Calculate total sqft; may need 2+ units	160 sqft coverage limitation per unit
Budget planning	Factor \$108/year ongoing Power Drops cost	Prevents subscription fatigue after purchase
Plant care	Maintain 50-70% humidity, indirect light	Optimal conditions ensure system effectiveness
Air quality testing	Use Neoplants' free air quality assessment	Establishes baseline to measure improvement

Sources

1. [Neoplants Official Website](#)
2. [Neo Px Product Page](#)
3. [AgFunder News - Market Analysis](#)
4. [CB Insights Company Profile](#)
5. [CB Insights Financials](#)
6. [CB Insights Competitors](#)
7. [Growjo Company Data](#)
8. [Neoplants Scientific White Paper](#)

